

Artificial Intelligence

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Learning Outcomes of Lecture 04

- Understand the important concepts relevant to the landscape of search
- Master the rationale of hill-climbing search
- Master the rationale of simulated annealing search
- Understand the important concepts relevant to genetic algorithm
- Master how to solve the problem using genetic algorithm

Agenda

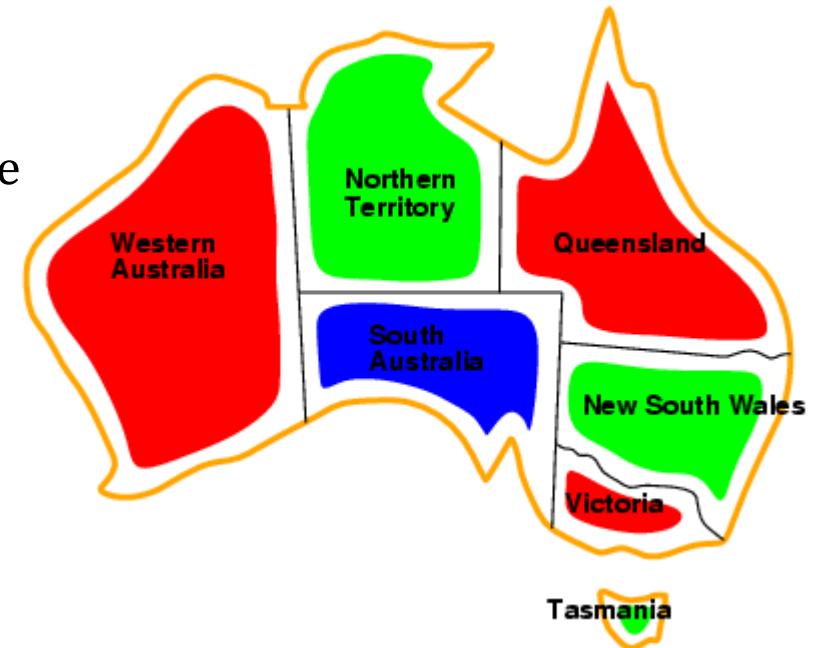
- Constraint
 - Something that controls what you do by keeping you within particular limits
- Constraint satisfaction problem
- Games

Agenda

- Constraint
- Constraint satisfaction problem
- Games

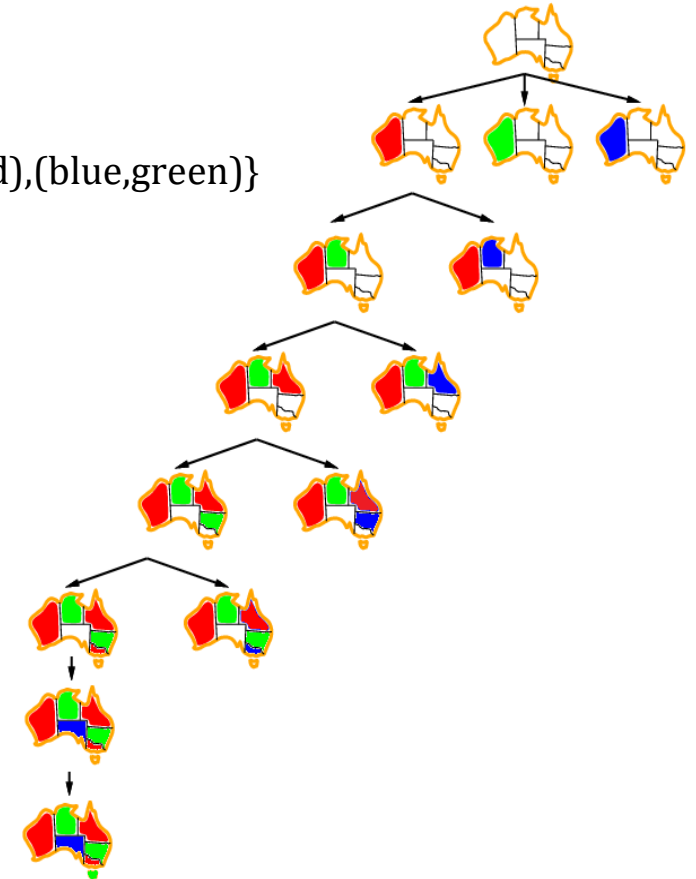
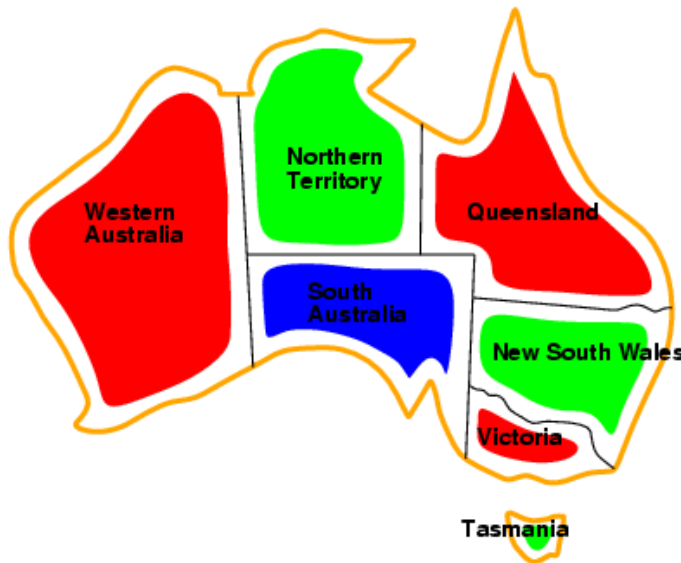
Constraint Satisfaction Problems

- Map-coloring
 - Adjacent regions must have different colors
 - WA = red, NT = green, Q = red, NSW = green, V = red, SA = blue, T = green
- Classical search problem
 - Each state is atomic with no internal structure
 - These states can be evaluated and test to see whether they are goal states
- Constraint satisfaction problem (CSP)
 - State is defined by variables x_i with values from domain D_i
 - Goal test is a set of constraints specifying allowable combinations of values for subsets of variables
- CSP consists of three components
 - A set of variables
 - A set of domains
 - A set of constraints



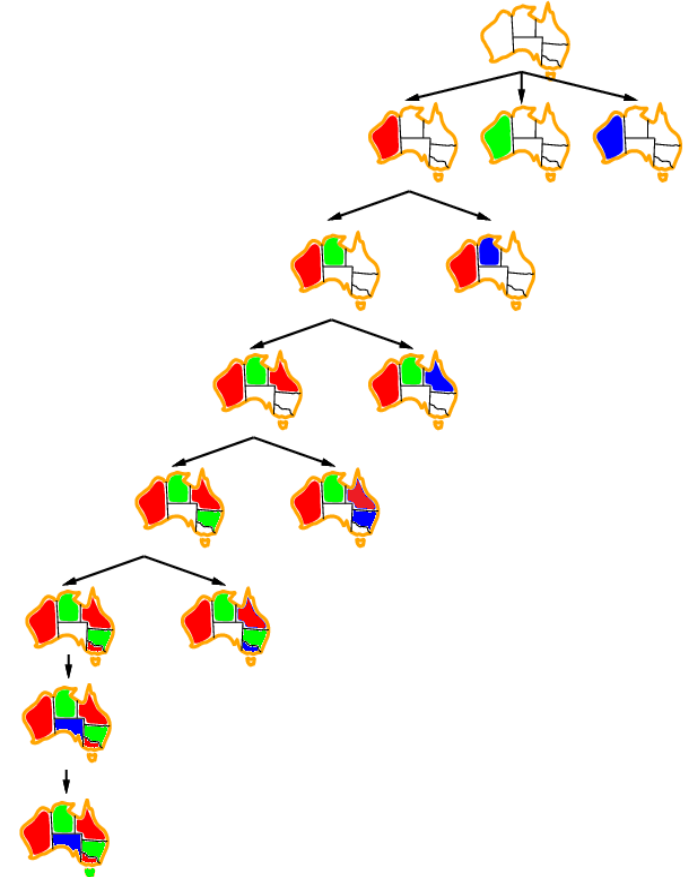
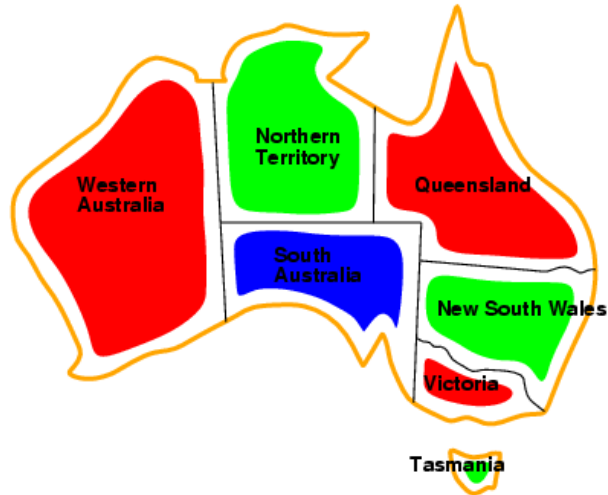
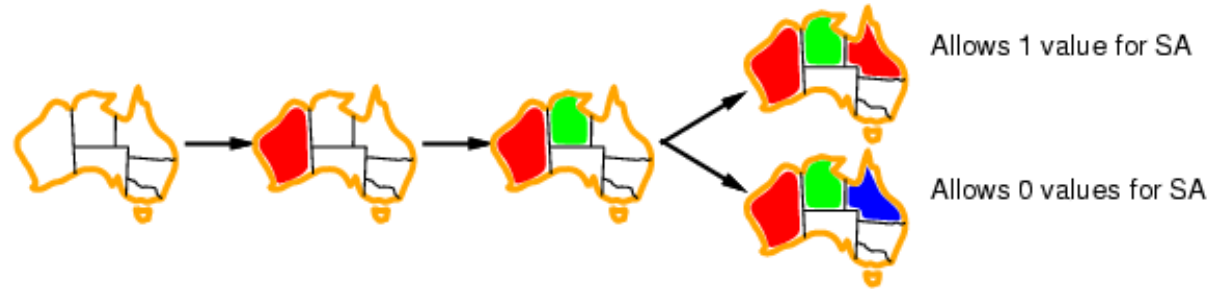
Solve Map-Coloring Problem using Search Tree

- Variables {WA, NT, Q, NSW, V, SA, T}
- Domains $D_i = \{\text{red, green, blue}\}$
- Constraints: adjacent regions must have different colors
e.g. $WA \neq NT$, or $(WA, NT) \in \{(\text{red, green}), (\text{red, blue}), (\text{green, red}), (\text{green, blue}), (\text{blue, red}), (\text{blue, green})\}$



Least Constraining Value

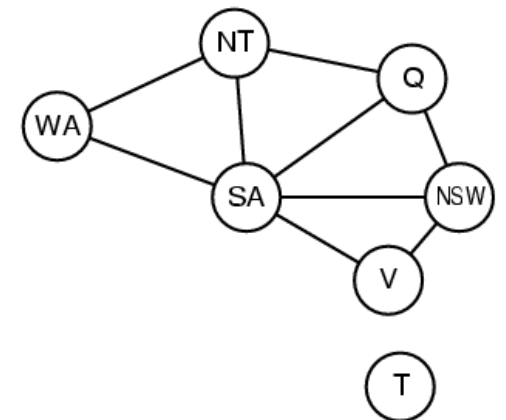
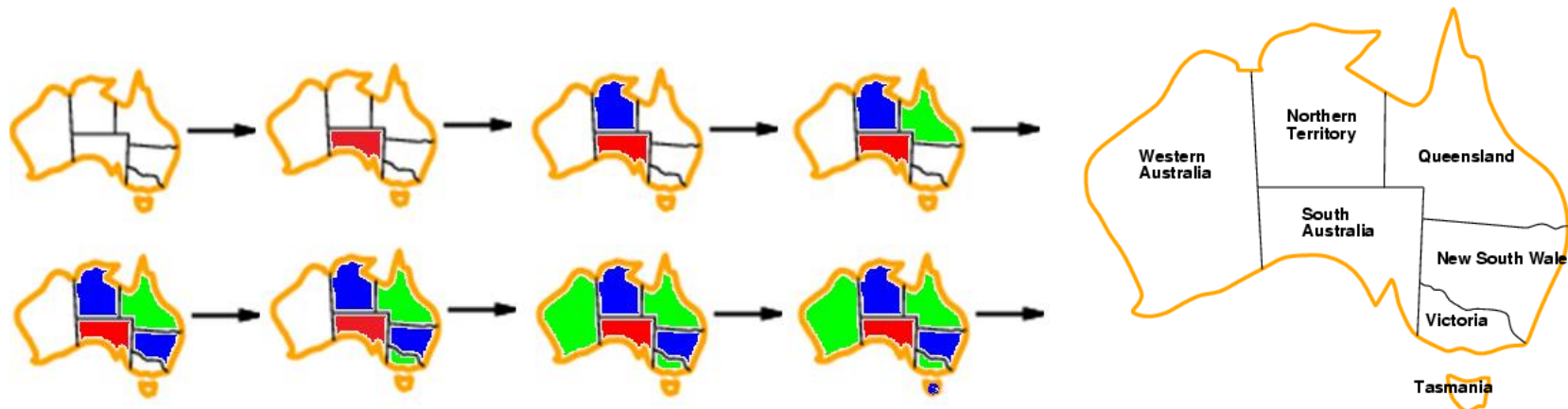
- Strategies to speedup the search
 - Given a variable, choose the least constraining value



Most Constrained Variable

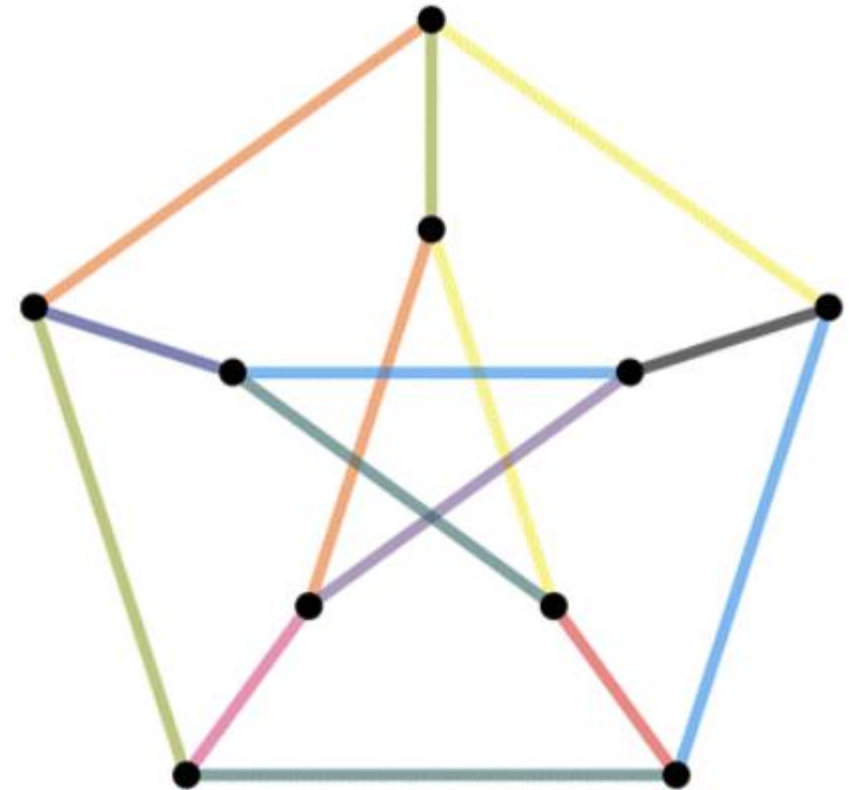
- Constraint graph
 - nodes are variables, arcs are constraints
- Most constrained variable
 - Choose the variable with the fewest legal values

Variable	WA	NT	SA	Q	NSW	V	T	Weight
WA	0	1	1	0	0	0	0	2
NT	1	0	1	1	0	0	0	3
SA	1	1	0	1	1	1	0	5
Q	0	1	1	0	1	0	0	3
NSW	0	0	1	1	0	1	0	3
V	0	0	1	0	1	0	0	2
T	0	0	0	0	0	0	0	0



Applications

- CSP enable solutions of complex problems
 - ❑ Exam and class scheduling
 - ❑ Job scheduling
 - ❑ Air traffic flow management
 - ❑ Bandwidth allocation in wireless communication
 - ❑ Register allocation in computer compiler



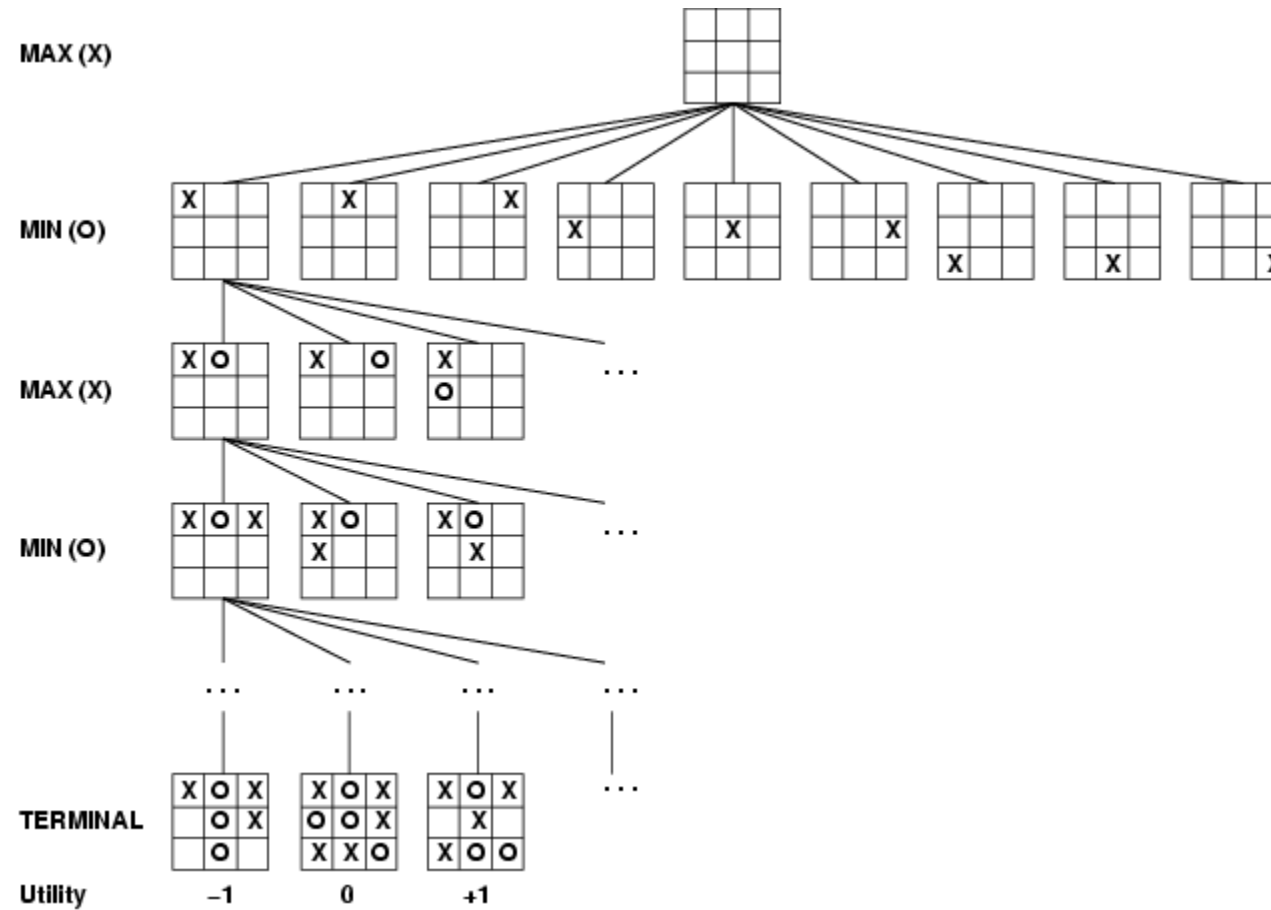
Agenda

- Constraint
- Constraint satisfaction problem
- Games

Games

- Multiagent environment
 - In which each agent needs to consider the actions of other agents
 - And how they affect its own welfare
- Games
 - Views any multiagent environment as a game, provided that the impact of each agent on the other is significant
- Games in AI
 - The state of a game is easy to represent
 - Agents are usually restricted to a small number of actions
 - Outcomes are defined by precise rules

Example of a Partial Game Tree

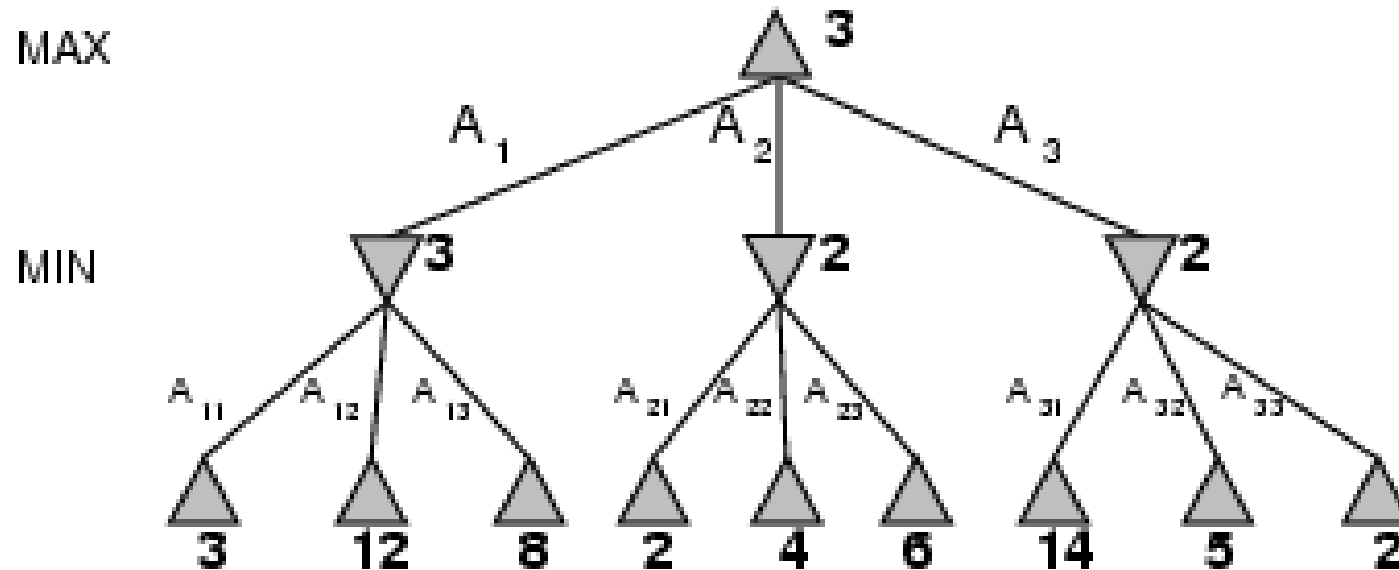


Formulate Game to Search Problem

- S_0
 - The initial state, which specifies how the game is set up at the start
- $\text{Player}(s)$
 - Defines which players has the move in a state
- $\text{Actions}(s)$
 - Defines a set of legal move in a state
- $\text{Result}(s, a)$
 - The transition model, which defines the result of a move
- $\text{Terminal-Test}(s)$
 - A terminal test, which is true when the game is over
- $\text{Utility}(s, p)$
 - A utility function, also called an objective function or payoff function
 - Defines the final numeric value for a game that ends in terminal state s for a player p

Optimal Decision in Games

- Perfect play for deterministic games
 - Idea: choose move to position with highest minimax value
 - best achievable payoff against best player



α - β Pruning

- α
 - Best already explored option along path to the root for maximizer
- β
 - Best already explored option along path to the root for minimizer

